

RDU Hourly Temperature Forecast

Course-flow technical walkthrough | Existing frozen experiment | No post-test reselection

Research story: seasonality is a strong baseline; OLS adds almost nothing; a recent anomaly adds a meaningful and interpretable improvement; HGB captures more nonlinear structure and scored lower error on this test, but is more complex and warm-biased. The conclusion is a performance-versus-interpretability/generalization tradeoff.



1. Problem

One forecast is issued at 2026-09-17 00:00 EDT (04:00 UTC) for every hour through 2026-09-30 23:00 EDT: 336 hours with no new observations after issue time.

2. Data and cutoff

Public IEM routine ASOS/METAR reports provide RDU temperature plus RDU/GSO/CLT dew point, pressure, wind speed, and wind direction. Development begins 2015-01-01 and ends strictly before the 2026-09-17 04:00 UTC cutoff. Development and final-test directories are separate, and request/load guards enforce the cutoff.

3. Target construction

The implementation maps :45-:59 routine observations to the following nominal hour and keeps the latest eligible station-hour report. The actual timestamp remains available for audit. This is the project's operational course-target mapping; exact timestamp equivalence requires an official instructor label file. Invalid fields become missing and targets are never interpolated. Clean data: 307,323 station-hours; latest 14-day RDU development coverage: 100%.

4. Feature engineering

A causal seasonal Ridge reference uses earlier years only. Known calendar/lead features are combined with origin-known RDU anomalies (24/72/168 hours), coverage, age, range, trend, dew-point depression, pressure change, wind components, and GSO/CLT anomalies. Weather observed inside the future window is never a feature. Feature groups are compared by paired historical-validation ablation only.

Group	Examples	Availability
Calendar / lead	harmonics, interactions, 1-336	known future
RDU state	24/72/168h anomaly, age, range	origin-known only
Weather / neighbors	dew point, pressure, wind, GSO/CLT	past reports only

5. Baselines

Seasonal climatology averages the same hour in a nearby calendar-day window using training data only. Previous-day persistence repeats the last complete day. Seasonal baseline test RMSE is 7.654°F; persistence is 8.745°F.

6. Required linear regression

OLS uses calendar harmonics and interactions. Test RMSE is 7.653°F versus 7.654°F for the seasonal baseline: essentially no added final-test accuracy beyond seasonality.

7. Interpretable anomaly model

$\text{anomaly_tau72} = \text{causal seasonal reference} + \text{recent 72-hour anomaly} \times \exp(-\text{lead}/72)$. It reacts to recent warm/cool departures, then fades the correction. Validation RMSE 5.768°F; test RMSE 7.307°F; MAE 6.097°F; R^2 0.492; bias -0.272°F.

8. Nonlinear HGB comparison

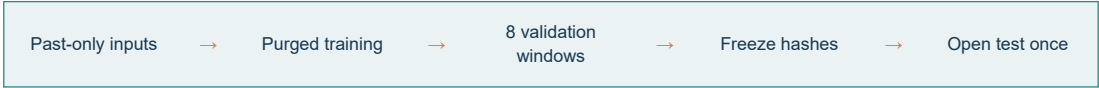
`hgb_neighbors_l15` learns a nonlinear correction from RDU state, local weather, neighbors, and lead. Validation RMSE 5.879°F; test RMSE 6.969°F; MAE 5.298°F; R^2 0.538; bias +3.386°F. It scored lower test error but is more complex and substantially warm-biased.

Model	Validation RMSE	Test RMSE	Test MAE	R²	Bias
<code>anomaly_tau72</code>	5.768°F	7.307°F	6.097°F	0.492	-0.272°F
<code>hgb_neighbors_l15</code>	5.879°F	6.969°F	5.298°F	0.538	+3.386°F

HGB achieved lower error on this final test, but the validation-selected primary model was not changed after test access.

9. Temporal validation

Eight 14-day windows: Sep 17 in 2021-2023; Aug 15 and Sep 17 in 2024-2025; Aug 15 in 2026. Each predicts all 336 hours. Training purges any trajectory crossing the fold origin. Imputation/scaling fit inside each fold; annual references use prior years. Development/test directories are separate. Source, data, models, and predictions are frozen before test access. Final scoring performs zero fitting or selection.



10. Evaluation metrics

RMSE = sqrt(mean squared error) and is primary because it penalizes large hourly misses and supplies one fixed validation criterion. MAE is average absolute error; bias is mean(prediction - observation); R² is explained variation relative to the period mean, not classification accuracy. All 336 final hours have authentic labels.

Metric	Purpose
RMSE	Primary; penalizes large misses
MAE	Typical absolute error
Bias	Warm/cool direction
R ²	Explained variation; not accuracy

11. Validation-only model selection

Nineteen fixed configurations are ranked by mean validation-window RMSE. The rule prefers lower complexity only within 1% of the minimum, then lower RMSE. anomaly_tau72 is the minimum at 5.768°F. HGB at 5.879°F is about 1.9% worse, outside the 1% near-tie region. Therefore anomaly wins on validation performance under the rule, not merely because it is simpler.

5.879 / 5.768 - 1 = 1.9%: HGB is outside the predefined 1% near-tie region.

12. Freeze

Step 06 fits the declared final comparison set, saves 336 predictions, and freezes code, data, model, and prediction hashes before step 07 opens final-test labels. The evaluator refuses to proceed if frozen evidence changes.

13. Final untouched test

Model	RMSE °F	MAE °F	Bias °F	R ²	Role
Seasonal baseline	7.654	6.317	-0.124	0.443	Baseline
OLS harmonics	7.653	6.398	-0.695	0.443	Required linear
anomaly_tau72	7.307	6.097	-0.272	0.492	Selected primary
hgb_neighbors_115	6.969	5.298	+3.386	0.538	Nonlinear comparison

This table is evaluation only, not reselection. The primary model remains anomaly_tau72.

14. Conclusions / tradeoffs / limitations

Seasonality carries most predictable structure. Recent anomaly offers the clearest useful improvement. HGB adds nonlinear performance on this test, with complexity and warm-bias costs. A station-only 14-day model cannot know future fronts, cloud, precipitation, or circulation. IEM may include retrospective corrections; the 15-minute availability buffer is an assumption. The final test is now published, so future changes require a new untouched test period.

15. Code map

Stage	Implementation
Configuration / integrity	config.py; integrity.py
Data / target	step01_download.py; step02_clean.py
Features	step03_features.py
Models	step04_models.py
Validation / selection	step05_backtest_select.py
Fit / freeze	step06_freeze_forecast.py
Final scoring	step07_final_evaluate.py
Visualization / entry	step08_visualize.py; 00_run_pipeline.py

Data source: Iowa Environmental Mesonet ASOS/METAR archive - <https://mesonet.agron.iastate.edu/request/download.phtml>